# [2024-2025 春夏] 离散数学理论基础 Quiz4

悔教 @cc98

2025 年 7 月 1 日

1. (20%) Determine whether the following statements are true or false.
   (5 points for a correct answer, 1 point for a blank answer, 0 points for an incorrect answer)

   a) $\forall x P(x) \wedge \exists x Q(x)$ and $\forall x \exists y \big(P(x) \wedge Q(y)\big)$ are logically equivalent. ( )

   b) $1+2=4$ if and only if $1-3=7$. ( )

   c) If a relation $R$ is not transitive then $R^2$ is also not transitive. ( )

   d) There is not any countable infinite set $A$ with a bijection $A \to A \times A$. ( )

2. (40%, 5% for each blank)

   (1) Suppose that a full 3-ary tree has 60 internal vertices. It has _____ leaves.

   (2) Suppose $S = \{1,2,3,4,5,6,7,8,9\}$. If $S$ has a subset $X$ such that the sum of its elements is odd, we call $X$ an odd subset (e.g., $\{8,9\}$). If $S$ has a subset $Y$ such that the sum of its elements is even, we call $Y$ an even subset (e.g., $\{2,4,8\}$). Which is greater: the number of odd subsets or the number of even subsets? Or are they equal? _____

   (3) If $G$ is a planar connected graph with 20 vertices, each of degree 3, then $G$ has _____ regions.

   (4) 10 balls are arranged in a row from left to right, including 4 indistinguishable red balls and 6 indistinguishable green balls. No two red balls are allowed to be adjacent. How many different arrangements are there? _____

   (5) 10 balls are arranged in a row from left to right, including 4 labeled red balls and 6 labeled green balls. No two red balls are allowed to be adjacent. How many different arrangements are there? _____

   (6) How many four-digit decimal numbers (1000–9999) are divisible by 3 and contain the digit 3?

   _____

   (7) Exactly how many trailing zeros are in the decimal form of 30!? _____

   (8) Arrange the following functions in increasing order such that each function is $O$ of the next one:

   $$\log(n^2+9n), \quad \log\log n, \quad n\log n, \quad \log 3^n, \quad n\cdot 2^n, \quad 9^n, \quad n^9, \quad (\log_{1.1} n)^{99}, \quad n!, \quad \log(n!)$$

   _____

3. (8%) Let

$$A = \{(a,b) \mid a,b \in \mathbb{Z}^+\}.$$

Define the relation $R$ on $A$ as

$$R = \{((a,b),(c,d)) \mid ad + a = bc + c\}.$$

(1) Determine whether $R$ is an equivalence relation.

(2) Prove your answer.

4. (9%) Solve the following recurrence relation with the given initial conditions:

$$a_n = 6a_{n-1} - 12a_{n-2} + 8a_{n-3} \quad (n \geq 3),$$

with

$$a_0 = -5, \quad a_1 = 4, \quad a_2 = 88.$$

5. (8%) There are 8 cards, each with a front and a back (making a total of 16 faces). The letters i, l, o, v, e, z, j, and u are each written exactly twice on these faces in any order. When arranging the cards in a row, each card may be placed with either side facing up. Prove that the 8 cards can be arranged in a row shows "ilovezju" .

6. (15%) In a complete graph $K_n$, each edge is colored with either red or blue. If two edges incident to a vertex (i.e., forming an angle) are of the same color, we call that a monochromatic angle. If three vertices form a triangle with all three edges of the same color, it is called a monochromatic triangle.

(1) Prove that for $n = 6$, there are at least 24 monochromatic angles.

(2) Prove that for $n = 6$, there are at least 2 monochromatic triangles.

(3) Prove that for $n = 7$, there are at least 2 monochromatic triangles of the same color.